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July 11, 1974
SUPERSEDING
Int. Fed. Spec. HH-T-0029
January 17, 1964

FEDERAL SPECIFICATION

TAPE, ADHESIVE, LEAD FOIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a lead foil adhesive tape designed for stop off masking during electroplating and where a moisture proof seal is required.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-680 - Tape, Pressure-Sensitive Adhesive: Packaging and Packing Of.

Federal Standard:

Fed. Test Method Std. No. 147 - Tapes, Packaging, Adhesive and Gummed, Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Construction. The tape shall consist of a lead foil backing, that is free of pin holes, and is coated on one side with a pressure-sensitive adhesive that requires no heat, moisture, or other preparation prior to application except that of removal of a liner that shall cover the face side of the adhesive. The liner shall be easily removed without transfer of the adhesive from the backing to the liner.

3.2 Rolls. The tape shall be wound in rolls, liner side in, on cores of sufficient rigidity to prevent distortion of the rolls under normal conditions of transportation, storage, and use. The inside diameter of the core shall be 3 minus 0, plus 1/16 inches. No more than three splices will be permitted in each roll of tape. Any splice shall be such that it will not separate when the roll is unwound in test or application.

3.3 Length and width. The tape shall be furnished in 36 yard long rolls. The width shall be as specified in the contract or order. A width tolerance of ±1/32 inch shall be allowed.

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3.4 Physical properties. The test values must meet the following requirements.

3.4.1 Adhesion. When tested as specifies in 4.4.2, the adhesion shall be 30 ounces per inch of width minimum.

3.4.2 Tensile strength. When tested as specified in 4.4.2, the tensile strength at break shall be 18 pounds per inch of width minimum.

3.4.3 Elongation at break. When tested as specified in 4.4.2, the elongation at break shall be 15 percent minimum.

3.4.4 Moisture vapor transmission rate. When tested as specified in 4.4.2, the moisture vapor transmission rate shall be 0.5 grams per 100 square inches per 24 hours maximum.

3.4.5 Thickness. When tested as specified in 4.4.3, the backing thickness shall be 0.0045 to 0.0055 inches.

3.5 Weathering. When tested as described in 4.4.4, the tape shall show no evidence of lifting and shall have a minimum adhesion of 35 ounces per inch of width.

3.6 Storage shelf life. When stored as described in 6.3, the useful shelf life shall be 12 months. Tape that has shelf life extended shall not be used for electroplating operations.

3.7 Workmanship. Workmanship shall be first class throughout. The tape shall be free from defects which detract from its appearance or impair its serviceability. The adhesive shall be a smooth, uniform coating, covering the entire area of one side of the tape. Edges of the tape shall be straight, true, and unbroken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 End item examination. Sampling and examination of the end item shall be as specified in section 7 of Fed. Test Method Std. No. 147.

4.3 End item testing. Sampling and examination for end item testing shall be as specified in section 7 of Fed. Test Method Std. No. 147 at an Acceptable Quality Level (AQL) of 4 percent.

4.4 Tests. Tests for acceptance of individual lots shall consist of tests for all requirements specified in section 3.

4.4.1 Test Conditions. Unless otherwise specified in the test method, the rolls of tape shall be conditioned for at least 24 hours at 73° ± 3.5° F. and 50 ± 2 percent relative humidity prior to testing and shall be tested under these conditions.

4.4.2 Test methods. The following tests shall be conducted as described in Fed. Test Method Std. No. 147, except that one specimen per sample unit shall be tested:

Adhesion to steel ----------Method 10.2
Tensile strength at break ----------Method 30.2
Elongation at break ----------Method 30.2
Moisture Vapor Transmission Rate----Method 70.2

4.4.3 Thickness. The thickness of the foil backing shall be determined as described in Method 36.2 of Fed. Test Method Std. No. 147 after the liner and adhesive have been removed.

4.4.4 Weathering. One specimen per sample unit shall be tested as described in Method 63.2 of Fed. Test Method Std. No. 147. Exposure shall be for 96 hours on a standard steel panel. Examine for conformance to 3.5. Test for adhesion as described in Method 10.2, see 4.4.2, except do not remove the tape from the test panel after weathering..

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The packaging of individual rolls of tape shall be as described in either 5.1.1 or 5.1.2.

5.1.1 Individual rolls shall be packaged in containers made of fiberboard conforming to PPP-B-636, type CF, class-domestic, and centering devices shall provided.

5.1.2 The packaging of individual rolls shall be in containers made of paperboard conforming to PPP-B-566. Containers conforming to variety 1, style II, type D, class C (pye lock inner flaps) shall be used for roll widths less than 3/4 inch. Containers conforming to variety 1, style II, type D, class E (locked inner flaps) shall be used for roll widths equal to or greater than 3/4 inch. A centering device formed from 0.025 to 0.030 inch linearly aligned polyethylene shall be used. The centering device shall have a friction fit core holding device (1/2 inch deep and 2-15/16 inches in diameter) and a 6 inch diameter flange with a 1/2 inch deep reverse lip. One centering device shall be used for roll widths less than or equal to 1 inch. Two centering devices (one on each side of the roll) shall be used for roll widths greater than 1 inch.

5.2 Packing. Individual packaged and multiple rolls shall be packed as specified in PPP-T-680. The quantity of rolls per shipping container shall be as follows:

Width of roll (inches)	No. of individual packaged rolls per shipping container	No. of Multiple rolls per shipping container
1/4	12	48
1/2	12	24
3/4	12	16
1	12	12
1-1/2	6	8
2	6	6
3		4
4		3

5.3 Marking. Containers of tape shall be marked in accordance with the requirements of PPP-T-680. When specified, the core of each roll of tape shall be marked on the inside or on the edge with numerals or letters indicating the month and year of manufacture and specification number. In addition, the manufacturer's name and designation of the product shall be marked on the core.

6. NOTES

6.1 Intended uses. This tape is intended for use for stop off masking during electroplating, especially in hard chrome plating where a "thieving action" is desired. It is also a pressure-sensitive tape with a very low moisture vapor transmission rate, and therefore, is ideally suited for sealing containers, building panels, and similar type applications demanding a semi-permanent moisture seal.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and data of this specification.
(b) Width of tape required (see 3.3).
(c) Selection of applicable levels of packing.

6.3 Storage. The tape should be stored in the original container in a cool location. It should not be stored in close proximity to steam pipes, radiators, or other sources of heat, or in a location where it may be exposed to direct rays of sunlight.

Preparing activity:
GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 25 cents each.

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